Comsats University Islamabad, Lahore Campus

Sessional II Examination – SPRING 2020

Course Title	Information Security	Course Code		CSC432	Credit Hours		3(2,1)
Instructor/s	Dr. Tariq Umer, Dr. M Sharjeel	Program Name		BSE/BCS			
Semester	7^{th},8^{th}	Batch	FA16	Section	All	Date	March 15, 2020
Time Allowed	1 hr 30 minutes	Maximum Marks		25			
Student Name:		Registration No.					

Important Instructions/Guidelines

1. Use of S-Box, Lookup table cheat sheet is allowed
2. Exchange of calculator is not allowed
3. Avoid cutting and overwriting.
4. Return the question paper along with the answer sheet. **Missing paper will result in elimination of 5 marks**

Question # 01 **[3]; CLO :<1>; Bloom Taxonomy Level: <*Understanding*>**

Identify the type of attack in the provided scenarios:

1. You receive a phone call. The caller claims that they are from a bank, and ask your debit card number and OTP for the purpose of account confirmation, otherwise your account will be blocked.
2. You tried to login to your flex account during which you ignored a warning displayed by the browser and later on you came to know that your login credentials have been stolen.
3. The outgoing network traffic of your system has suddenly increased and on diagnosis, you have identified that your system is constantly sending a specific request to a server.

Question # 02 **[2]; CLO :<1>; Bloom Taxonomy Level: <*Remembering*>**

Suppose an app is heavily used at your workplace. The app was written a long time ago in C language. Unfortunately the app source code was lost due to a hard disk failure. Your boss is concerned about some buffer overflow vulnerabilities that may exist in the app. What (technical) actions would you recommend to prevent buffer overflow exploits?

Question # 03 **[2*5]; CLO :< 3,4>; Bloom Taxonomy Level: <*Analyzing*>**

1. Why parallel encryption of blocks in not possible in Cipher block chaining (CBC).
2. Differentiate between known plain text attack and chosen plain text attack.
3. Apply PLAYFAIR cypher of depth 4 on "COMSATS LAHORE".
4. How cryptology is different from cryptography?
5. Illustrate the Fiestel structure with the help of a diagram.

Question # 04 **[5*2]; CLO :<4>; Bloom Taxonomy Level: <*Applying*>**

1. Compute the output of the *Mix Columns* transformation for the following sequence of input bytes “67 89 AB CD”
2. Generate 3rd round key from the previous key for AES:

7e	**05**	**00**	**10**
55	**20**	**a3**	**03**
3f	**4d**	**22**	**78**
10	**08**	**2a**	**20**

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16